CYBER SECURITY REVIEW

"Social Engineering Tactics and Their Impact on Security"

Group 2

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PREFACE

First of all, thanks to Allah SWT, who has given us the opportunity to complete this task. Blessing and salutation may also be upon the most honorable prophet Muhammad SAW who has given us life guidance from our birth to the end of life

This paper explores the various social engineering tactics used by attackers, such as phishing, pretexting, and baiting, and examines their growing impact on cybersecurity. By reviewing these techniques and their consequences, the paper aims to highlight the importance of user awareness and the need for stronger human-centric defenses in modern security strategies.

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Authors

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CHAPTER I

INTRODUCTION

I.1 Background

In the digital age, cyber security has become a major concern for individuals, organizations, and governments alike. As technology continues to evolve, so do the methods used by cybercriminals. One of the most effective and commonly used methods is social engineering, a type of attack that targets people rather than systems. Unlike traditional cyberattacks that rely on technical weaknesses, social engineering exploits human behavior, such as trust, curiosity, fear, or urgency, to trick victims into giving away confidential information or access.

These tactics have proven to be highly effective because even the most secure systems can be compromised if the people using them are not property informed or cautious. Social engineering attacks, like phishing emails or impersonation scams, often bypass technical defenses, making them a serious threat in today's cybersecurity landscape. Understanding how these attacks work and their overall impact is essential in building stronger security awareness and creating more effective strategies to prevent them.

So that's why this cyber security review has been made. to give a reader a full concept and explanation about social engineering from all aspects, mainly the method and the prevention.

I.2 Writing Objectives

The main objective of this paper is to provide a comprehensive review of social engineering tactics and analyze their impact on cybersecurity. This includes identifying the most common techniques used by attackers, understanding how these tactics exploit human behavior, and evaluating the risks they pose to individuals and organizations.

I.3 Problem Domain

Because technological developments continue to develop, we must also anticipate negative things that are also developing. In this paper we explain social engineering, studying how attackers play with psychology to gain the privacy of the victim

in this Cyber security review, it focused on social engineering attacks, from the general basic knowledge, types of the attack, way to prevent, and etc. these will be the main topics of the Cyber security review

The issues the authors discussed in this Cyber security review are:

- Definition of social engineering
- Types of social engineering
- Prevention
- Research Methodology

I.4 Writing Methodology

Our writing method is descriptive analysis. Data research techniques such as descriptive analysis help in the description, illustration, or summarization of information points, allowing the development of patterns that fulfill all the requirements of the data. through online resources and user feedback

I.5 Writing Framework

The writing framework for this paper will consist of four chapters, starting with the introduction of this research paper. And then we'll dive into the basic theory, and then we'll explain about further details of the subject. And then finishing off with a conclusion and suggestion.

Chapter I: Introduction This chapter contains background, writing objectives, problem domain, writing methodology, and writing framework

Chapter II: Basic Theory This chapter will explain more about the general things about Phishing itself

Chapter III: Problem Analysis In this chapter, the explanation of Phishing will be more in depth by presenting all the details

Chapter IV: Conclusion and Suggestion This chapter contains conclusions and suggestions regarding the research we are conducting and observe

CHAPTER II

BASIC THEORY

II.1 Definition of Social Engineering

In cybersecurity, the term social engineering refers to psychological manipulations to someone with the objective of getting their private information, valuables, or access into something. Instead of exploiting technical vulnerabilities, social engineering targets human behavior and decision-making to achieve its objectives.

Social engineers rely on people's interactions to carry out their attacks. They often take advantage of online services and platforms to set up more advanced attacks, with the goals of gaining access to steal sensitive information or break into protected computer networks.

The use of communication technologies, the advancement of technologies, and the internet in both private and public settings has made the problem significantly worse. For example, the most common social engineering attacks are telephone calls, emails, and face-to-face interactions. Social engineering attacks often involve impersonating a known trustworthy company, such as banks.

II.2 Types of Social Engineering Attacks

Social engineering attacks ignore all the hardware and software that server has the purpose of preventing attacks in general, instead it targets the users directly. Social engineering is more challenging to do if the user is protected by both good hardware and software security systems. To prevent social engineering attacks, we must involve the person using the device, not just relying on technology like firewalls or antivirus software.

Social engineering attacks are diverse and change depending on the use of the technique. The following explains the various types of social engineering attacks.

a. **Phishing**

The attacker sends messages that have been modified so that targets believe that the messages received are legitimate from trusted organizations such as banks. Usually the targets are required to follow the instructions or suggestions in the message.

b. Grooming

This method is one of the newest techniques in social engineering, where the attacks uses psychological techniques and information technology to obtain confidential information related to potential victims of pedophilia. This method uses communications technology like SMS, emails, and telephone. The conversations i smore of a trap to become a victim of pedophilia.

c. Pretexting

In a pretexting attack, the attacker gathers publicly available information such as details from websites, social media, or phone directories, to build a believable story. They use this story to trick the victim into sharing confidential information. The attacker often pretends to offer help, a job opportunity, or a prize, or may ask to confirm personal details, making the interaction seem legitimate and trustworthy.

d. Baiting

This technique works by taking advantage of employees who are naturally curious. Attackers use something that sparks their interest to trick them into taking unsafe actions. The attackers use electronic media in physical form, such as USB drives that have been infected with malware or trojans.

e. Face to Face Interaction

Attacker usually use this method to achieve private information or file, how's this work is the attacker take the advantage of the victim's psychological weakness, such as helping or debugging any error and seduce them to get the physical access

F. Scareware or pop up windows

This attack uses a pop up window or browser that appears on a website while the user is doing any activities. certain web pages have been inserted codes that can run a pop ups that can be adjusted to target victims, such as ads, fake virus detector, and etc

G. Profile Cloning

This method uses the availability of information in public, such as social media, the attacker acts as someone whose profile is being cloned to get crucial information from other people. the public information can be in images, videos, full name, profile picture, even the way they talk from forum or comment. from all this resources the attacker will try to find a way to get the personal information, start to contacting the victim relation to get any important information

II.3 Prevention for Social Engineering Attacks

Hatfield's research presents the concept of social engineering attacks; it states that the best way to prevent them is to educate and train potential victims [1]. Accordingly, victims must have sufficient knowledge to defend against social engineering attacks. Deploying a health campaign is one of the prevention strategies for social engineering attacks. Identifying a social engineering attack as a semantic attack is challenging because of the behavioral deception that technical defenses cannot detect [12]. and the solution for social engineering attacks prevention are

1. **Social engineering policy**

Khidzir et al. [40] Establish management policies for social engineering prevention at four levels. At level one, risk management is carried out against social engineering attacks. At level two, the responsibilities and procedures for carrying them out are described. There is a list of evaluations for preventing social engineering attacks at level three, complete with instructions on how an activity is expressly said to have been completed. At level four, there is a framework that can be implemented into the system to prevent social engineering attacks and is able to accommodate digital evidence of social engineering attacks.

2. **Prevention protocol**

The prevention protocol, known as the co-utile disclosure or private data, could manage and protect private data that is shared on social networks, it works on two steps, the first step it evaluates the privacy risks of sharing files, based on the data and structure. the second step it calculates a privacy functionality score, which measure the benefit a user gets from sharing files and the risk of exposing private file

3. User Studies

Any journal, and research paper. can be used to learn knowledge on any social engineer attacks, the way to prevent it, and how to profile users who are more vulnerable. these research base on quantitative and qualitative methods

4. Personality and Behavioral processing

Cai et al. [51] created a cooperative data sanitization method that alters user profiles and friend connections on social networks. This helps protect users from social engineering attacks—especially inference attacks that try to uncover private details by analyzing public data. Their method makes it harder for attackers to accurately guess sensitive user information.

CHAPTER III

PROBLEM ANALYSIS

III.1 Research Methodology

Based on the journal the participants were selected from different organizations, the research sample is divided into two groups. The first group contained participants who had knowledge of social engineering approaches, while the other group contained participants who had no prior knowledge of social engineering practices.

Organizations chosen based on the dependence of their business on information technology and the risk level of a social engineering attack, and an equal percent employee obtained from each organization. Men and women participated equally in this research.

In order to identify the level of awareness of social engineering attacks, the researcher made a scale for the level of social engineering knowledge, the scale is weak, moderate, and good knowledge. The question is grouped into four categories, the categories are knowledge, practices, solutions, and education.

Table 1. Level of awareness of social engineering.

Awareness level	**Frequence**	**Percent**
Weak knowledge	214	42.1%
Moderate knowledge	256	50.4%
Good knowledge	38	7.5%
Total	**508**	**100.0%**

Table 2. Pearson correlation for social engineering.

Items	**Correlation**
10) Do you have knowledge that there is an attack on your device?	0.567**

11) Do you know how to deal with it if there is an attack on your computer or a virus?	0.640**
12) Do you have knowledge about the cybercrime system?	0.604**
18) Is the USB considered a transferor for viruses?	0.436**
20) Is there an anti-virus's software in your device?	0.636**
21) Are you updating your anti-virus software regularly?	0.720**
23) Is the cost of the ani-virus program appropriate?	0.551**
24) Are you updating your operating system usually?	0.549**
25) Have you ever taken a course in Social Engineering?	0.459**
26) Do you want to take some courses about Social Engineering?	0.406**
**** : Correlation is significant at the 0.01 level (2-tailed)**	

Table 3. Item-total statistics for social engineering.

Item	**Total Correlation**	**Cronbach's Alpha**
10) Do you have knowledge that there is an attack on your device?	0.420	0.736
11) Do you know how to deal with it if there is an attack on your computer or a virus?	0.503	0.724
12) Do you have knowledge about the cybercrime system?	0.468	0.729
18) Is the USB considered a transferor for viruses?	0.303	0.751

20) Is there an anti-virus's software in your device?	0.507	0.723
21) Are you updating your anti-virus software regularly?	0.605	0.707
23) Is the cost of the ani-virus program appropriate?	0.392	0.741
24) Are you updating your operating system usually?	0.414	0.737
25) Have you ever taken a course in Social Engineering?	0.337	0.747
26) Do you want to take some courses about Social Engineering?	0.234	0.763

III.2 Result

According to the journal, the total number of participants in this study was 508 people, with a response rate of 99%, including 382 (75.2%) male and 126 (24.8%) female as it illustrated in Figure 1. 151 (29.7%) participants were considered to be at the age of 18 - 25 years old, while 145 (28.5%) were deemed to be around the age of 26 - 35 years old.

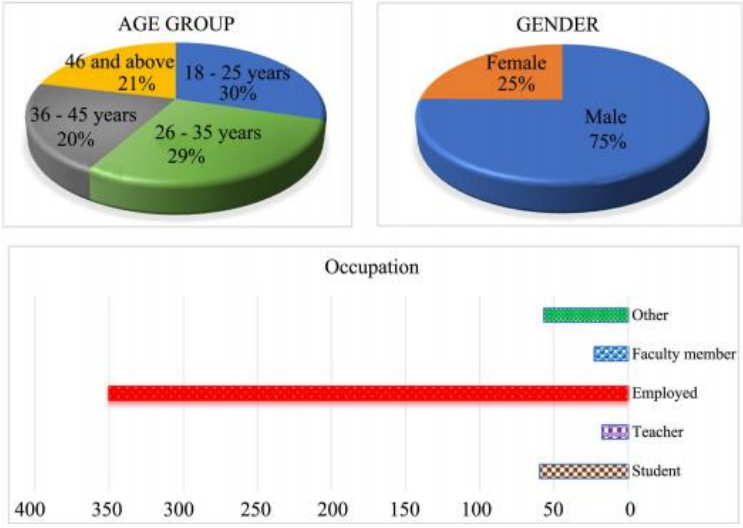

Figure 1. Characteristic of the participants according to demographic variables; june 2022

Table 4. Correlation between knowledge about social engineering & the demographic characteristics.

Variable		Do you know what social engineering		P-Value
		No, I have no idea	**Yes, I do**	
Age Group	18 - 25 years	89 (27.6%)	62 (33.3%)	0.167
	26 - 35 years	90 (28.0%)	55 (29.6%)	
	36 - 45 years	65 (20.2%)	39 (21.0%)	
	46 and above	78 (24.2%)	30 (16.1%)	
	Total	**322 (100.0%)**	**186 (100.0%)**	
Gender	Male	247 (64.7%)	135 (35.3%)	0.299
	Female	75 (59.5%)	51 (40.5%)	
	Total	**322 (100.0%)**	**186 (100.0%)**	
Occupation	Student	41 (12.7%)	19 (10.2%)	0.080
	Teacher	11 (3.4%)	7 (3.8%)	
	Administrator	r 220 (68.3%)	130 (69.9%)	
	Faculty Member	9 (2.8%)	14 (7.5%)	
	Other	41 (12.7%)	16 (8.6%)	
	Total	**322 (100.0%)**	**186 (100.0%)**	
What are the usual purposes	Course Registration	20 (6.2%)	14 (7.5%)	0.504

of using the internet	Paying school fees	4 (1.2%)	1 (0.5%)	
	shopping	16 (5.0%)	6 (3.2%)	
	chatting	29 (9.0%)	12 (6.5%)	
	any other purpose	82 (25.5%)	41 (22.0%)	
	all of the above	171 (53.1%)	112 (60.2%)	
	Total	**322 (100.0%)**	**186 (100.0%)**	
Do you know any type of social engineering threats	No, I have no idea	308 (95.7%)	34 (18.3%)	**0.000****
	Yes, I do	14 (4.3%)	152 (81.7%)	
	Total	**322 (100.0%)**	**186 (100.0%)**	

The study showed that many people lack awareness of social engineering. Based on the results in Table 1, only 7.5% of participants had good knowledge of social engineering, while the majority—42.1%—had weak knowledge, and 50.4% had just moderate knowledge. This shows a clear gap in understanding that could leave many individuals and organizations at risk.

When the researchers looked at how different knowledge areas relate to each other (shown in Table 2), they found strong connections. For example, people who knew how to deal with attacks or regularly updated their antivirus software had higher overall awareness. One of the strongest relationships was with the question about updating antivirus software regularly, which had a high correlation of 0.720, meaning that people who did this were also likely to be more informed about social engineering.

The consistency of these results was backed up by a reliability test (in Table 3). This test showed that all the items measured in the questionnaire were trustworthy, with Cronbach's Alpha values ranging between 0.707 and 0.763. This means the questions were well-designed and the data collected is dependable.

Looking deeper at demographics in Table 4, the study found that knowledge levels were not strongly affected by age, gender, or occupation—except when it came to knowing specific types of social engineering threats. There was a very significant difference here: 95.7% of people who didn't know what social engineering was also didn't know any specific threats, while 81.7% of those who did know about social engineering could identify threats. This shows that prior knowledge clearly boosts awareness.

III.3 Discussion

The journal aim is to improve an understanding of the levels of awareness of social engineering approaches in organizations. The researcher of this journal conducted this study based on the dependence of their business information technology and the risk level of a social engineering attack. The researcher developed the questionnaire according to social engineering knowledge.

Table 5. LSD-Post Hoc Test ANOVA (occupation) the study group (social engineering awareness); 2022

Dependent Variable	(I) Select your age group	(J) Select your age group	Mean Difference (I-J)	Sig.
Need for education courses	36-45 years	18-25 years	0.085	0.031
		26-35 years	0.073	0.069
		46 and above	0.120*	0.005

Table 6. T-test (gender) for the study group versus the (social engineering awareness), Saudi Arabia; 2022

Gender		**N**	**Mean**	**SD.**	**F**	**t**	**Sig.**
Social	Male	**382**	**1.624**	**0.286**	**0.345**	**−2.006**	**0.045***

engineering and information security knowledge	Female	**126**	**1.683**	**0.276**			
Information security practices	Male	**382**	**1.872**	**0.231**	**7.303**	**0.435**	**0.664**
	Female	**126**	**1.862**	**0.191**			
Technical security solutions	Male	**382**	**2.339**	**0.526**	**4.147**	**−2.327**	**0.020***
	Female	**126**	**2.462**	**0.485**			
Need for education courses	Male	**382**	**1.390**	**0.300**	**0.341**	**−1.829**	**0.068**
	Female	**126**	**1.448**	**0.340**			

The study found that the general awareness of social engineering among organizational members is low, with most participants showing only weak or moderate knowledge. This is concerning because social engineering attacks rely on manipulating people, and without proper awareness, individuals are more likely to fall victim.

The researchers discovered that prior knowledge significantly improves cybersecurity practices. Those who had taken courses or were familiar with social engineering concepts were better at identifying threats and using security tools like antivirus software. This was confirmed by strong statistical results in earlier tables and was further supported by the analysis of demographic differences.

According to Table 8, people aged 36–45 were the most in need of educational courses on social engineering, especially when compared to younger participants. Table 9 showed that females had significantly better awareness in certain areas, particularly in terms of general knowledge and use of technical security tools. Table 10 revealed that faculty members were more interested in further education, while administrators scored lower in cybersecurity practices.

These findings suggest that targeted training programs—based on age, gender, and job roles—can be more effective. Organizations should focus on providing regular awareness sessions, especially for groups like administrators and employees in the 36–45 age range, who were shown to be more vulnerable.

CHAPTER IV

CONCLUSION AND SUGGESTION

IV.1 CONCLUSION

The Conclusion of this Cyber Security review paper is to be more open in the digital era, and be more cautious. Because social engineering is a psychological attack, they use human error as a vulnerability. and this report consist of the explanation, types, prevention, and etc

Always taking care of our private information because we know social engineer attacks are always evolving, so our knowledge prevents it. always pay attention to any suspicious activity on the internet

IV.2 SUGGESTION

Our suggestions as an author from this paper are. Always pay attention to who is emailing us, suspicious links, and fake senders. always take care of your own private information and credentials. only trust the verified and secure institution if you have to give the private information.

If the reader wants to know more about this social engineering, and the related things. You can find a lot of information on the internet such as journals, research papers, and youtube videos.

and also pay attention to the journal you want to read, the journal that has been standardized. like scopus,IEEE, and etc.

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